Emergency Department Wait Time Prediction: Systematic Literature Review

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Introduction to Literature Review

Why emergency department (ED) wait time prediction has become an operational machine learning problem

Problem background

EDs worldwide face severe overcrowding and growing wait times. Prolonged wait times worsen patient outcomes, increase mortality, reduce staff efficiency, and drive patients to leave without being seen.

ML methods leverage triage variables, patient demographics, and hospital operational data to estimate wait time and optimize patient flow.

Predictive systems matter because hospitals can use early forecasts to optimize staffing, improve patient routing, and reduce crowding before queues worsen.

Core logic across the literature

All reviewed studies use the same modeling principle: combine patient severity with system pressure information

Common variables

- Triage level (ESI/MTS)
- Patient demographics
- Arrival time & patterns
- ED workload & crowding
- Queue status
- Service-flow measures

Differs

- Prediction of the outcome
- Data source / hospital
- Same operational goal

Research Question and Themes

The literature review organizes the project around one prediction objective and four recurring themes

Research Question: How can supervised ML algorithms trained on routinely collected ED triage data accurately predict patient wait times, and what methodological approaches produce the most clinically interpretable and generalizable models?

Why the question matters

ED wait time is not only a patient experience metric. ED wait time is also a queue-management signal that affects crowding, throughput, staffing, and timelines of care. A useful model therefore needs both patient-level features and system-level features.

1 Wait time prediction
Supervised models estimate wait time or prolonged wait from triage and queue variables.

Binary & Regression Targets

2 AI-based triage
Text-based and triage-based models enrich early patient assessment.

NLP + Structured Data

3 Resource optimization
Forecast support staffing, room assignment, and congestion response.

Operational Decision Support

4 Real-time forecasting
The hardest challenge is producing predictions useful during live operations.

Deployment Challenge

Most recurrent predictors
Triage level
Patient demographics
Arrival time and temporal patterns
Current emergency departments workload
Queue and service-flow variables

Interpretation: The literature constantly treats waiting time as a joint function of patient severity and operational congestion- both dimensions are needed for useful prediction

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Paper 4: Benchmarking Emergency Department Predictions

Xie, F., Zhou, J., Lee, J. W., et al. **(2022)**

Research focus

First open-source ED prediction benchmark using the publicly available MIMIC-IV-ED dataset, enabling reproducible comparisons across models and institutions.

Methodology

- Dataset: MIMIC-IV-ED: (~448k visits, 2011-2019).
- Models: LR, RF, Gradient Boosting, MPL Med2Vec, LSTM, Autoscore.
- Features: Triage vitals, ESI level, demographics, ICD comorbidities.
- Tasks: hospitalization, critical outcomes (ICU/mortality), 72-hr reattendance.

Key findings

Gradient Boosting achieved AUROC 0.880 for critical outcomes. RF & GB outperformed all clinical scoring systems (MEWS, NEWS, REMS, CART). Deep learning showed only marginal improvement over classical ML with low-dimensional ED data. AutoScore balanced interpretability and accuracy.

Strengths

- First large-scale public ED benchmark (400+ visits).
- Open-source pipeline on GitHub (reproducible).
- Published in Nature Scientific Data.
- Enables model-to-model comparison across studies.

Limitations

- Does not address wait time prediction directly.
- Single institution (Beth israel Deaconess, Boston).
- No SHAP or interpretability analysis.
- No ESI triage-level segmentation or fairness analysis.
- Single institution.

Why the paper matters for our project:
Uses our exact dataset (MIMIC-IV-ED) and provides the open-source preprocessing pipeline we build on. Baseline model AUROCs serve as direct benchmarks. We add continuous wait time regression with SHAP and ESI subgroup analysis as a fourth prediction task.

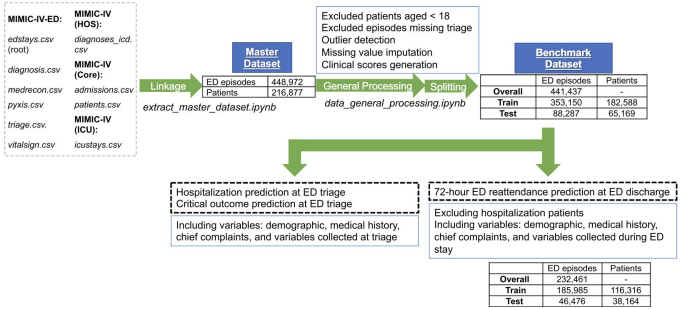


Fig. 2 The workflow of data processing from raw data.

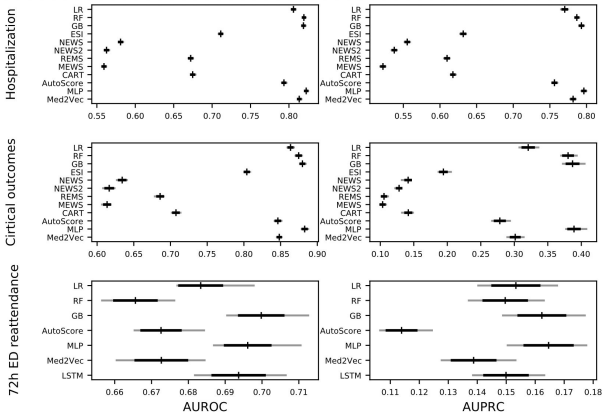


Fig. 3 Bar plots comparing the performance of various prediction models based on three different outcomes. AUROC: The area under the receiver operating characteristic curve. AUPRC: The area under the precision-recall curve. CART: Cardiac Arrest Risk Triage. ESI: Emergency Severity Index. GB: Gradient boosting. LSTM: Long short-term memory. LR: Logistic regression. MEWS: Modified Early Warning Score. MLP: Multilayer perceptron. NEWS: National Early Warning Score. NEWS2: National Early Warning Score, Version 2. REMS: Rapid Emergency Medicine Score. RF: Random forest.

Xie, F., et al. (2022). Benchmarking emergency department prediction models with machine learning and public electronic health records. *Scientific Data, 9*, 658. https://doi.org/10.1038/s41597-022-01782-9

Paper 7: Interpretable Machine Learning Models for Prolonged Emergency Department Wait Time Prediction

Wang, H., Sambamoorthi, N., Sandlin, D., and Sambamoorthi, U. **(2025)**

Research focus

Comprehensive analysis of ML models to predict prolonged ED wait time (≥30 min) in ESI-3 patients, with global and local SHAP interpretation of key features and feature interactions.

Methodology

- Dataset: 177,665 ESI-3 visits, Fort Worth TX (Jan 2019 - Dec 2021).
- Models: CVLR, Random Forest, XGBoost, ANN, SVM (70/30 split, 5-fold cv).
- Target: binary - prolonged wait time (≥30 min vs <30 min.
- Metrics: Accuracy, recall, precision, F1, AUROC, FRP, FNR. SHAP beeswarm + PDP used for interpretation).

Key findings

~48% of patients prolonged wait time. All five models showed comparable accuracy (~0.75). Minimizing FNR was most clinically relevant. Top predictors: ED crowding status and mode of arrival (ambulance). Female patients and Hispanic/Latino patients had higher prolonged wait times.

Strengths

- Only study comparing 5 ML algorithms specially for ED wait time.
- SHAP beeswarm + PDP + feature interaction analysis.
- Large real-world single-center dataset (177K+ visits).
- Clinical framing: FNR prioritized over FPR.

Limitations

- Binary classification only - no continuous wait time regression.
- ESI-3 patients only - not generalizable to all triage levels.
- Single-center study; limited external validity.
- No formal demographic fairness mitigation proposed.

Why the paper matters for our project:

Directly mirrors our use case with same prediction target (ED wait time), same model family (XGBoost, RF, and same interpretability tool (SHAP). Provides a direct methodological template and performance baseline. We extended to continuous regression, all 5 ESI levels, and MIMIC-IV-EDfor full reproducibility.

Table 2 Performance accuracy comparison of using five different machine learning algorithms with all features to predict patient wait times

	CVLR		Random Forest		XGBoost		ANN		SVM	
	Training	Testing	Training	Testing	Training	Testing	Training	Testing	Training	Testing
Accuracy	0.75	0.74	0.76	0.74	0.75	0.75	0.75	0.74	0.75	0.75
Recall	0.80	0.80	0.80	0.79	0.79	0.79	0.80	0.79	0.80	0.80
Precision	0.71	0.71	0.73	0.71	0.71	0.71	0.72	0.71	0.71	0.71
F1 score	0.75	0.75	0.76	0.75	0.75	0.75	0.75	0.75	0.75	0.75
AUROC	0.81	0.81	0.81	0.81	0.81	0.81	0.82	0.81	0.79	0.78

The performance accuracy was reported with the use of training and testing data

Abbreviations: *XGBoost* eXtreme Gradient Boosting, *CVLR* Cross Validation Logistic Regression, *ANN* Artificial Neural Network, *SVM* Support Vector Machine, *AUROC* Areas Under the Receiver Operating Characteristics

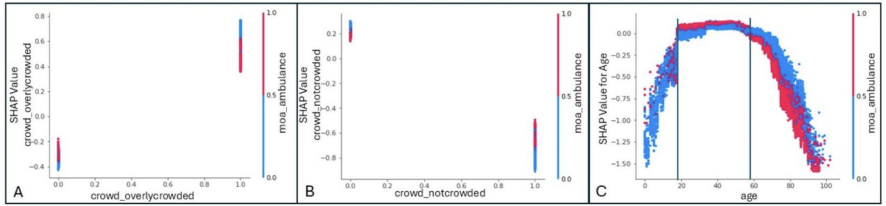


Fig. 5 Partial dependency plots of top feature interactions using XGBoost algorithm for prolonged wait time predictions. Figure 5 shows the PDP of top feature interactions. **A** shows the interactions between ED overly crowded conditions and ambulance transportation. When ED was overly crowded, more patients with ambulance transfer had less prolonged wait times than patients who arrived at ED without ambulance transfer. **B** shows when ED was not crowded, more patients who arrived without ambulance transfer had less prolonged wait times than patients who arrived via ambulance. **C** shows the interactions between age and ambulance transfer. Two perpendicular lines were drawn and indicated that patients' age ranges (20 to 60). Generally, less prolonged wait time occurred when patients aged either younger than 20 or older than 60. ED wait times were quite similar among patients aged ranging from 20- to 60-year-old. When patients older than 60, more patients had less prolonged ED wait times when arrived at ED via ambulance than ones without

Wang, H., Sambamoorthi, N., Sandlin, D., & Sambamoorthi, U. (2025). Interpretable machine learning models for prolonged emergency department wait time prediction. *BMC Health Services Research, 25*, 403. https://doi.org/10.1186/s12913-025-12535-w

Paper 9: Development of an Emergency Department Length-of-Stay Prediction Model Based on Machine Learning

Wu, W., Li, M., Jiang, H., Sun, M., Zhu, Y., Zhu, G., Li, Y., Li, Y., Mo, J., Chen, X., and Mao, H. **(2025)**

Research focus

Develop and compare ML regression models to predict continuous Emergency Department length of Stay (EDLOS) in minutes, identifying the most accurate algorithm and key predictive factors.

Methodology

- Dataset: 187,028 adult ED visits, Guangzhou Medical University (Mar 2019 - Sep 2021).
- Models: LR, LASSO, Ridge, GBDT, XGBoost, LightGBM.
- Target: Continuous EDLOS (minutes).
- Evaluation: MAE, MSE, RMSE, R-squared (cross-validated; 70/30 split). Feature importance via LightGBM.

Key findings

LightGBM achieved the best performance: MAE = 443.5, RMSE 826.8, R-squared = 0.48 - significantly outperforming linear baseline (R-squared = 0.37). Top 3 predictors: emergency wait time (feature importance 0.219), age (0.169), emergency arrival time (0.091) Triage level, diagnosis, and arrival date also ranked in top 10.

Strengths

- One of few ED using continuous regression (minutes) rather than binary classification..
- Large real-world dataset (187K cases).
- Directly validates LightGBM and XGBoost as strong regression baselines for EDLOS.
- Clear LightGBM feature importance ranking provided.

Limitations

- Single-center, single-country study - limited generalizability.
- No SHAP for patient-level interpretation.
- No ESI triage-level subgroup analysis.
- No fairness or demographic disparity analysis.
- Excludes unstructured EHR Text (chief complaint, history).

Why the paper matters for our project:
Validates the regression approach and confirms lightGBM/XGBoost as strong baselines for continuous EDLOS prediction. We replicate this regression framework on the public MIMIC-IV-ED dataset and extend it with SHAP interpretability, ESI-level segmentation, and a fairness assessment - addressing all four gaps identified in this study.

Table 2. Prediction performance of machine learning model

Models	MAE	MSE	RMSE	R^2
LR	552.7	826,918.416	909.351	0.371
LASSO	554.425	832,412.452	912.366	0.367
Ridge	566.901	830,210.089	911.159	0.368
XGBoost	466.703	712,896.776	844.332	0.458
GBDT	471.608	710,281.862	842.782	0.460
LightGBM	443.519	683,570.072	826.783	0.480

LR: linear regression; LASSO: least absolute shrinkage and selection operator; Ridge: ridge regression; GBDT: gradient boosting decision tree; XGBoost: eXtreme gradient boosting; LightGBM: light gradient boosting machine; MAE: mean absolute error; MSE: mean squared error; RMSE: root mean squared error; R^2: R-squared.

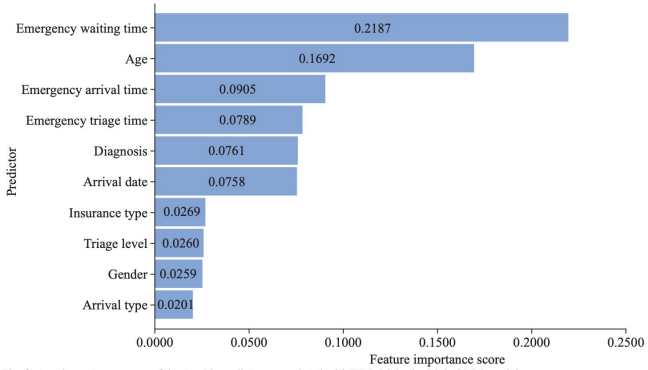


Figure 1. The feature importance score of the top 10 predictors associated with EDLOS in the LightGBM model.

Wu, W., Li, M., Jiang, H., Sun, M., Zhu, Y., Zhu, G., Li, Y., Li, Y., Mo, J., Chen, X., & Mao, H. (2025). Development of an emergency department length-of-stay prediction model based on machine learning. *World Journal of Emergency Medicine, 16*(3), 220–224. https://doi.org/10.5847/wjem.j.1920-8642.2025.048

Paper 12: ED-Copilot: Reduce Emergency Department Wait Time Language Model Diagnostic Assistance

Sun, L., Agarwal, A., Kornblith, A., Yu, B., and Xiong, C. **(2024)**

Research focus

Can a biomedical LLM combined with RL agent recommend sequential lab tests to reduce ED wait time while maintaining diagnostic accuracy? The only paper shifting from prediction to active clinical decision support.

Methodology

- Dataset: MIMIC-ED-Assist (from MIMIC-IV-ED), - 32,356 hospitalized ED visits; 12 lab groups, 68 tests.
- Model: BioGPT(345M params) + PPO reinforcement learning agent for sequential lab test selection.
- Targets: Critical outcome (death/ICU transfer in 12h) and LOS >24h.

Key findings

ED-Copilot halved average laboratory wait time from ~4 hours to ~2 hours while outperforming all tabular ML baselines. F1 for critical outcome: 0.413 (vs 0.394 best baseline). AUROC: 0.820. Personalized recommendations improved outcomes especially for high -severity (rare lab) patients. Bio-medical pre-training and RL were both necessary for strong performance.

Strengths

- Only paper applying LLM + RL for active ED lab ordering optimization.
- Published at ICML 2024 (top-tier ML venue).
- Uses MIMIC-IV-ED - same data source as our project.
- Demonstrates personalized, patient-specific recommendations.

Limitations

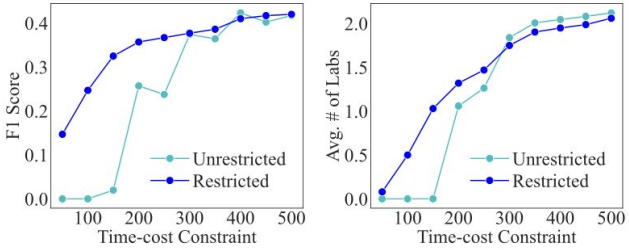
- Dataset biased to hospitalized patients only - not representative of full ED population.
- Binary prediction targets only - no continuous wait time regression.
- Computationally expensive (345M-parameters model + RL)..
- Single-center (Beth Israel Deaconess); no fairness or demographic analysis.
- Offline retrospective benchmark - no real-world clinical validation.

Why the paper matters for our project:

Demonstrates that reducing lab-ordering time is a meaningful lever for reducing ED LOS. We apply a simpler, interpretable continuous regression approach from triage data alone, adding all ESI levels, SHAP interpretability, and fairness analysis - with far lower computational cost.

Table 3. Ablation study on components of ED-Copilot, including linearization techniques, feature importance, and PLM backbones.

Group	Variations	Critical Outcome					Lengthened ED Stay				
		F1	AUC	Sensitivity	Specificity	Avg. Time-cost	F1	AUC	Sensitivity	Specificity	Avg. Time-cost
	ED-Copilot (345M)	**0.413**	**0.820**	**0.750**	**0.779**	**125 Min**	**0.252**	**0.707**	**0.725**	**0.606**	**154 Min**
Linearization	Raw Lab Test Name	0.397	0.777	0.768	0.677	134 Min	0.241	0.695	0.611	0.701	144 Min
Features	w/o. Triage	0.277	0.704	0.679	0.649	—	0.145	0.593	0.532	0.606	—
	w/o. CBC	0.385	0.803	0.692	0.777	—	0.224	0.686	0.696	0.596	—
	w/o. CHEM	0.420	0.827	0.788	0.746	—	0.234	0.702	0.656	0.606	—
Backbone	BioGPT (345M) w/o. RL	0.381	0.810	0.725	0.765	265 Min	0.236	0.718	0.710	0.620	265 Min
	Llama (7B LORA) w/o. RL	0.397	0.798	0.692	0.767	265 Min	0.232	0.701	0.705	0.610	265 Min
	Pythia (70M) w. RL	0.290	0.698	0.574	0.702	166 Min	0.178	0.596	0.555	0.619	126 Min
	GPT2-Medium (345M) w. RL	0.358	0.757	0.621	0.747	133 Min	0.166	0.539	0.498	0.584	96 Min



(a) Accuracy (b) Avg. # of Labs

Figure 5. Comparison of prediction performance on critical outcome of ED-Copilot when restricted or not restricted to tests patients performed in its suggestion at different time-cost constraints.

Sun, L., Agarwal, A., Kornblith, A., Yu, B., & Xiong, C. (2024). ED-Copilot: Reduce emergency department wait time with language model diagnostic assistance. In *Proceedings of the 41st International Conference on Machine Learning*(Vol. 235, pp. 46942–46956). PMLR. https://proceedings.mlr.press/v235/sun24a.html

Research Gaps Identified

The literature shows progress, but the literature also leaves several unsolved implementation problems

Limited Continuous Regression
Most studies use binary classification (prolonged vs. not) Only articles 9 and 10 use continuous targets, No study applies regression specifically to ED wait time prediction.

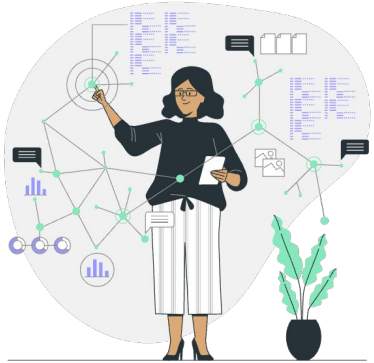
No Triage-Level Segmentation
No study reposts model performance broken down by ESI levels 1-5 separately. ESI-1 and ESI-4 patients have fundamentally different wait time dynamics.

Limited SHAP Interpretability
Only 2 of 10 empirical studies (articles 3 and 7) use SHAP. Article 8 uses TE2Rules as an alternative. Most studies report only global feature importance or none at all.

Proprietary Datasets
7 of 10 empirical studies studies use private single-center hospital data. Only articles 4, 11, and 12 use MIMIC-IV-ED (public), limiting reproducibility and cross-institutional comparison.

Incomplete Fairness Evaluation
Only article 3 and 11 address fairness - article 3 identifies disparities but proposes no mitigation; article 11 test debiasing strategies but focuses on admission, not wait time.

Retrospective Models Only
Most models are retrospective research artifacts rather than live operational tools. Real-time deployment challenges remain largely unaddressed in the literature.



These gaps directly motivate our project:
Regression-based ML + SHAP + ESI segmentation +fairness on public MIMIC-IV-ED data

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How Our Projects Build Upon Existing Work

The literature shows progress, but the literature also leaves several unsolved implementation problems

ED triage data collection (MIMIC-IV-ED)

↓

Data cleaning & feature engineering

↓

Model training (XGBoost, RF, Gradient Boosting)

↓

Model evaluation (MAE, RMSE, R-squared)

↓

SHAP-based feature importance analysis

↓

ESI-level subgroup & fairness analysis

Proposed approach

- Input data.
- Patient demographics.
- Triage level.
- Arrival time.
- Emergency department workload

Models:

- Random Forest.
- Gradient Boosting.
- XGBoost.

Project contribution vs Literature

- Continuous regression (minutes) - not binary classification.
- All 5 ESI triage levels - not just ESI-3.
- SHAP at patient + population level
- Public MIMIC-IV-ED - fully reproducible.
- Formal fairness assessment across demographics.

Research questions and preliminary expectation

Research question 1: Which combination of patient variables and operational variables best predicts wait time?

Research question 2: Which model balances predictive performance and interpretability most effectively?

Preliminary expectation: Ensemble models will perform strongly, but operational usefulness will depend on feature transparency and live deployment feasibility.

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Summary & Conclusion

What the literature establishes and what the project will do next

Key insights

- ED overcrowding remains a major global healthcare challenge affecting patient outcomes and staff efficiency.
- ML models - especially gradient boosting (XGBoost, LightGBM, RF) - consistently outperform traditional clinical scoring systems.
- Combining patient-level features (triage, demographics) with system-level features (crowding, arrival mode) improves prediction.
- Real-time deployment, interpretability, and cross-hospital validation remain the most under-addressed gaps in the field.
- The MIMIC-IV-ED public dataset has emerged as the reproducibility gold standard for ED prediction research

Our project contribution

- Build a predictive ML model for ED wait time.
- Use patient data and hospital operational data together.
- Support better resource allocation and operational decision-making.
- Apply SHAP analysis to identify key features driving wait time predictions.
- Assess model fairness across ESI triage levels and patient demographic subgroups.

Next steps

1. Dataset collection
2. Data cleaning and feature engineering
3. Model training and evaluation
4. Comparison of Random Forest, Gradient Boosting, and XGBoost
5. Interpretation of feature importance and operational usefulness

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